

Week 2 Master Guide.md

THE FINAL EXAM TRAP LIST (Review Before Exam)

- **The Temperature Trap: Molarity** depends on temperature (because liquid volume expands). **Molality** does *not* (because mass is constant). Also, when using the Ideal Gas Law ($PV = nRT$), **T** MUST be in **Kelvin**, not Celsius.
- **The Solids Sample Volume Trap:** If a question asks for TDS in mg/L and gives you TS = 250 mg/L, TSS = 190 mg/L, and a sample volume of "50 mL," **ignore the 50 mL!** It is a deliberate distractor. Just subtract: $250 - 190 = 60 \text{ mg/L}$.
- **The "N as N" Trap:** Reporting nitrogen species as "mg N/L" allows scientists to directly sum different forms (Ammonia, Nitrite, Nitrate) on a common mass basis to find Total Nitrogen, completely bypassing individual molecular weight conversions.
- **The Atmospheric Abundance Trap: Nitrogen (N_2)** and **Oxygen (O_2)** are the most abundant gases in the atmosphere, but they are **NOT** greenhouse gases.

EVERY Week 2 Concept (Explained Simply + Deep Facts)

Part 1: Concentrations & Chemical Units

- **1. Molarity vs. Molality:**
 - **Molarity (M):** Moles of solute per *Liter of solution*. Because liquid volume expands/contracts with heat, Molarity is **temperature-dependent**.
 - **Molality (m):** Moles of solute per *kg of solvent*. Because mass never changes with heat, Molality is **temperature-independent**.
- **2. Equivalent Weight & Normality (Formula Sheet Check):**
 - **Equivalent Weight:** The weight of a molecule divided by its "valency" or "n-factor" (how many hydrogen ions it can replace).
 - **Formula:** $\text{Eq. Wt.} = \text{MW} / n$
 - **Examples:** H_2SO_4 (MW=98, $n=2$) $\rightarrow 49 \text{ g/eqv.}$ Na_2CO_3 (MW=106, $n=2$) $\rightarrow 53 \text{ g/eqv.}$ HCl (MW=36.5, $n=1$) $\rightarrow 36.5 \text{ g/eqv.}$
 - **Normality (N):** Equivalents of solute per Liter of solution.
 - **Formula:** $N = M \times n$
 - **Example:** 1M of $H_2SO_4 = 2N$.
 - **Note:** Normality depends on both **Dilution** and **Temperature**.

- **3. The “Parts Per” Rules (ppm & ppb):**
 - **In Water (Aqueous):** $1 \text{ ppm} \approx 1 \text{ mg/L}$. (Valid for dilute aqueous systems where density is $\sim 1,000 \text{ g/L}$).
 - **In Soil/Solids:** $1 \text{ ppm} = 1 \text{ mg/kg}$ (mass/mass).
 - **In Air (Gases):** ppm_v is a **volume/volume** ratio.
- **4. Unit Classifications (Slide Concepts):**
 - **Fundamental Units:** Mass, length, time.
 - **Basic Units:** Fundamental units PLUS temperature, electrical current, luminous intensity, etc.
 - **Derived Units:** Area, volume, velocity (combinations of basic/fundamental).
- **5. Mass-Basis PPM Formula:** $\text{ppm} = \frac{m_i}{m_{\text{total}}} \times 10^6$. (Often tested as a conceptual or numerical question).

Part 2: Air, Gases & The Atmosphere

- **5. The Ideal Gas Law:** Used to convert gas concentrations between ppm_v and mass-based units ($\mu\text{g}/\text{m}^3$).
 - *Formula:* $P V = nRT$. (Temperature must be in **Kelvin**, not Celsius).
- **6. Dalton’s Law of Partial Pressures:** The pressure a specific gas exerts is directly proportional to its **mole fraction** (or volume fraction) in the mixture.
 - *Formula:* $P_i = y_i \times P_{\text{total}}$.
- **7. Atmospheric Gas Concentrations (Must-Know Trivia):**
 - **Nitrogen (N_2):** 780,840 ppmv (78.08%). *Most abundant, NOT a greenhouse gas.*
 - **Oxygen (O_2):** 209,460 ppmv (20.9%). *NOT a greenhouse gas.*
 - **Argon (Ar):** 9,340 ppmv (0.934%).
 - **Carbon Dioxide (CO_2):** 400 ppmv (0.04%). *Greenhouse gas.*
 - **Methane (CH_4):** 1.79 ppmv (or 1,790 ppbv). *Lowest partial pressure of this group; greenhouse gas.*
- **Ion Balance Error Acceptability:** Water must be electrically neutral. The analysis is checked via the formula: $E\% = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100$. If the error is $< 5\%$, the analysis is chemically acceptable.

Part 3: The Solids Family Tree & Water Chemistry

- **8. The Solids Breakdown:**
 - **TS (Total Solids):** Everything left after evaporating an unfiltered sample at 103°C . ($\text{TS} = \text{TSS} + \text{TDS}$).
 - **TSS (Total Suspended Solids):** The physical chunks caught on a filter, dried at 103°C . (Causes turbidity).

- **TDS (Total Dissolved Solids):** The invisible dissolved stuff (like salt) that passes through the filter, dried at 103°C. (*Note: High turbidity does NOT mean high TDS.*)
- **9. Volatile vs. Fixed Solids (The 550°C Fire Test):**
 - **VSS (Volatile Suspended Solids):** The fraction of TSS that burns off in a muffle furnace at 550°C. Represents **organic matter** (living/dead biomass).
 - **FSS (Fixed Suspended Solids):** The ash/mineral fraction that remains after 550°C. Represents **inorganic** material.
- **10. Hardness vs. Alkalinity:**
 - **Alkalinity Ions:** Anions (negative) like Bicarbonate (HCO_3^-) and Carbonate (CO_3^{2-}).
 - **Hardness Ions:** Cations (positive) like Calcium (Ca^{2+}) and Magnesium (Mg^{2+}).
- **11. Ion Balance Error:** Water must be electrically neutral. The acceptability is checked by:
$$\text{E}\% = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100.$$
 (Must be < 5% to be acceptable).

Part 4: Instruments, Statistics & Slide Secrets

- **12. Environmental Lab Instruments:**
 - **High Volume Sampler:** Sucks in huge amounts of air through a filter to measure **Particulate Matter** in the atmosphere.
 - **Chromatographs (GC / HPLC):** Sorts and separates complex gas or liquid mixtures into individual chemical components.
 - **Spectrometers (ICP / AAS):** Used to detect heavy metals and cations (like lead or arsenic) in water.
- **13. Statistical Data Handling (Audit Additions):**
 - **MDL (Method Detection Limit):** The minimum concentration a machine can measure with **99% confidence**.
 - **BDL (Below Detection Limit):** Statistically handled by assuming the value is zero, the MDL, or **1/2 the MDL**.
 - **Log-Normal Distribution:** Environmental data usually has a long tail (log-normal). We use the **Geometric Mean** as the measure of central tendency.
 - **UCL (Upper Confidence Limit):** A conservative estimate of the mean that takes into account uncertainties in the data set (e.g., 95% UCL).
- **14. Wet vs. Dry Weight:** To find the dry concentration of a pollutant, divide the wet concentration by the dry fraction (e.g.,

$$0.6 \text{ mg/kg wet} / 0.2 \text{ dry fraction} = 3.0 \text{ mg/kg dry}.$$

The Ultimate Week 2 Q&A Bank (Core + Spin-offs + Slides)

Units, Molarity, & Chemistry

- **Q:** The unit $\mu\text{g}/\text{m}^3$ is most appropriately used for reporting pollutant concentrations in: **A:** Air.
- **Q:** Among the following units which isn't a fundamental unit? **A:** Temperature (It is considered a 'Basic' unit, whereas Mass/Length/Time are 'Fundamental').
- **Q:** Molarity differs from molality because molarity depends on: **A:** Temperature and volume.
- **Q: Relationship between Molarity (M) and Normality (N):** [Slides]
 - **A:** $N = M \times n$ (where n is the valency or n-factor).
- **Q: Calculate the Normality (N) of 0.05 M H_2SO_4 .** [Slides]
 - **A:** 0.1 N. (Calculation: $0.05 \text{ M} \times 2 = 0.1 \text{ N}$).
- **Q: Why is Molality (m) preferred over Molarity (M) for precision?** [Slides]
 - **A:** Molality is temperature-independent (mass doesn't change with heat), whereas Molarity changes as liquid volume expands/contracts.
- **Q:** Why is Molality temperature-independent while Molarity is not? **A:** Molality divides by the *mass* of the solvent (constant), while Molarity divides by the *volume* of the solution (expands/contracts with temperature).
- **Q:** Which statement about ppm is correct? **A:** 1 mg of solute in 1 kg of solvent = 1 ppm.
- **Q:** When is the approximation $1 \text{ mg/L} \approx 1 \text{ ppm}$ valid? **A:** In most dilute *aqueous* (water) systems (density $\sim 1,000 \text{ g/L}$).
- **Q:** How is ppm expressed in the air? **A:** ppm_v = volume/volume ratio (not mass/mass).
- **Q:** What is the equivalent weight of H_2SO_4 ? **A:** 49 g/eqv.
- **Q:** What is the normality (N) of 1 M Solutions of H_2SO_4 ? **A:** 2N.
- **Q:** What is the equivalent weight of Na_2CO_3 ? **A:** 53.
- **Q:** What is the normality of a 1M solution of HCl? **A:** 1 N.
- **Q:** What is the molarity of 50 grams of calcium carbonate (CaCO_3)? **A:** 0.5 mol/L.
- **Q:** Calculate the molality of 30.0 g of NaCl dissolved in 500.0 mL of water. **A:** 1.03 m.
- **Q:** Convert 5 ppm of TCE (MW = 131.5) in water to Molarity. **A:** $3.8 \times 10^{-5} \text{ M}$.
- **Q:** What are the two factors that Normality depends on? **A:** Dilution and Temperature.
- **Q:** Reporting nitrogen as mg N/L instead of mg/L of individual species is useful because it: **A:** Accounts for all nitrogen forms on a common basis.
- **Q:** A mineral water analysis gives total cations = $1.0 \times 10^{-3} \text{ eqv/L}$ and total anions = $0.95 \times 10^{-3} \text{ eqv/L}$. Is the analysis acceptable? **A:** Yes (Ion Balance Error is $\sim 2.56\%$, which is $< 5\%$).
- **Q: What is the equivalent weight of Sodium Hydroxide (NaOH)?** [v2 Guide Spin-off]

- **A: 40 g/eqv.**
- **Calculation:** $MW = 40$. Since it has one OH^- group, $n = 1$.
Eq.Wt. = $40/1 = 40$.
- **Q: Calculate the Ionic Strength (I) of 0.01 M $CaCl_2$.** [Slides]
 - **A: 0.03.**
 - **Calculation:** $I = 0.5 \sum (c_i z_i^2)$.
 - $Ca^{2+} : 0.01 \times (2^2) = 0.04$.
 - $Cl^- : 0.02 \times (1^2) = 0.02$.
 - $I = 0.5 \times (0.04 + 0.02) = 0.03$.
- **Q: In the Ion Balance Error formula, what units must be used for the summation of Cations and Anions?** [Slide Veto]
 - **A: meq/L (milliequivalents per liter).**
 - **Note:** You cannot use mg/L directly; you must convert concentrations to equivalents first.
- **Q: The unit $\mu g/m^3$ is most appropriately used for reporting pollutant concentrations in:**
 - **A: Air.** (Mass per volume units are standard for compressible gases).
- **Q: When converting gaseous concentrations between ppm_v and $\mu g/m^3$, which law is fundamentally required?**
 - **A: Ideal gas law.**
- **Q: In a gas mixture, the partial pressure of a component gas is proportional to its:**
 - **A: Mole (or volume) fraction.**
- **Q: A mineral water analysis gives total cations = 1.0×10^{-3} eqv/L and total anions = 0.95×10^{-3} eqv/L. Is the analysis acceptable?**
 - **A: Yes.** (The Ion Balance Error is approx 2.56%, which is below the 5% threshold).

Gases, Pressures & Atmosphere

- **Q: Nitrogen (N_2) is the most abundant gas in the atmosphere. According to the slides, what is its primary role regarding atmospheric pressure?** [Slide Veto]
 - **A: It acts as a “diluent” for oxygen and provides the bulk of the atmospheric pressure.**
- **Q: What is partial pressure?** **A:** The pressure one of the gases in a mixture would exert if it alone occupied the whole container.
- **Q: In a gas mixture, the partial pressure of a component gas is proportional to its:** **A:** Mole (or volume) fraction.

- **Q:** What is the partial pressure of CO₂ if its concentration is 420 ppmv and atmospheric pressure is 1 atm? **A:** 4.2×10^{-4} atm.
- **Q:** What is the formula to calculate parts per million by volume (ppmv) from partial pressure? **A:** $\text{ppm}_v = (P_i/P_{\text{total}}) \times 10^6$.
- **Q:** When converting gaseous concentrations between ppmv and $\mu\text{g}/\text{m}^3$, which law is fundamentally required? **A:** Ideal gas law.
- **Q:** The ideal gas law uses T in degrees Celsius. **A:** False (It uses absolute temperature in Kelvin).
- **Q:** Which of the following is not a greenhouse gas? **A:** Oxygen (or Nitrogen).
- **Q:** Which gas exerts the lowest partial pressure in the atmosphere out of O₂, N₂, Ar, and CH₄? **A:** Methane (CH₄).

Water Solids & Ions

- **Q: Alkalinity in water is primarily caused by which three ions?** [Slides]
 - **A:** Bicarbonates (HCO_3^-), Carbonates (CO_3^{2-}), and Hydroxides (OH^-).
- **Q:** Total solids of a water sample are composed of total suspended solids and total dissolved solids. **A:** True.
- **Q:** TDS Calculation: TS = 250, TSS = 190, FSS = 60, VS = 130 mg/L. Find TDS. **A:** 60 mg/L.
- **Q:** How is TDS actually measured in the lab? **A:** Place the *filtrate* (water that passed through the filter) in a crucible and dry it at 103°C for 24 hours.
- **Q: In solids analysis, what is the relationship between TSS, VSS, and FSS?** [Slides]
 - **A:** $\text{TSS} = \text{VSS} + \text{FSS}$. (Total Suspended Solids = Volatile + Fixed).
- **Q:** Why does a highly turbid water sample not necessarily indicate high TDS? **A:** Turbidity is caused by *suspended* particles (TSS). Dissolved solids pass right through filters and do not cause cloudiness.
- **Q:** Volatile suspended solids (VSS) are primarily an indicator of: **A:** Organic matter content.
- **Q:** What is the difference between FSS and VSS? **A:** FSS is the inorganic mineral fraction left after 550°C ignition. VSS is the organic fraction that burns off.
- **Q:** Which ions are responsible for Alkalinity in water? **A:** CO_3^{2-} (Carbonate) and HCO_3^- (Bicarbonate).
- **Q: Which instrument is used to measure the “True Color” of a water sample after filtration?** [2023 Assignment]
 - **A:** Spectrophotometer (at 455 nm).

Instruments & Statistics

- **Q: When environmental data are log-normally distributed, which measure of central tendency is used?** [Slides]

- **A:** Geometric mean
- **Q:** Particulate matter present in atmosphere is measured by which instrument? **A:** High volume sampler.
- **Q:** Which instrument separates complex gas or liquid samples into single components? **A:** Gas Chromatograph (GC) or HPLC.
- **Q:** Which instrument analyzes heavy metals and cations? **A:** Inductively Coupled Plasma (ICP) or Atomic Absorption Spectrometer (AAS).
- **Q:** The Method Detection Limit (MDL) represents: **A:** The minimum concentration measurable with 99% confidence.
- **Q:** If you have sample results Below the Detection Limit (BDL), what are the options for handling them statistically? **A:** Assume them to be zero, assume them to be the detection limit, or assume them to be 1/2 the detection limit.
- **Q:** When environmental data are log-normally distributed, the most appropriate measure of central tendency is the: **A:** Geometric mean.
- **Q:** What does the Upper Confidence Limit (UCL) represent in environmental data? **A:** A conservative estimate of the mean that takes into account uncertainties in the data set.

ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: What is the difference between 1M HCl and 1M H₂SO₄ in terms of Normality?

***Answer:** Because HCl only has one hydrogen ion ($n = 1$), $1M = 1N$. Because H₂SO₄ has two replaceable hydrogen ions ($n = 2$), its normality is doubled, meaning $1M = 2N$.*

Quiz 2: You have a water sample that is highly turbid (cloudy). Does this automatically mean it has a high Total Dissolved Solids (TDS) concentration?

***Answer:** No. Turbidity is caused by Total Suspended Solids (TSS). Dissolved solids pass through the filter and are invisible, so they do not directly cause turbidity.*

Feynman Challenge: Explain the difference between **Molarity** and **Molality** to a 10-year-old, and tell them why a scientist in a hot desert might prefer to use Molality.

***Example Answer:** Imagine making a glass of lemonade. Molarity is measuring how much lemon powder is in the whole glass of liquid. Molality is measuring how much lemon powder you mixed with an exact weight of plain water. If you go to a hot desert, the liquid lemonade expands and takes up more space in the glass, so the*

Molarity changes. But the weight of the water never changes no matter how hot it gets, so Molality stays exactly the same!